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Nanotribology of hydrogels with similar stiffness but different polymer and crosslinker concentrations

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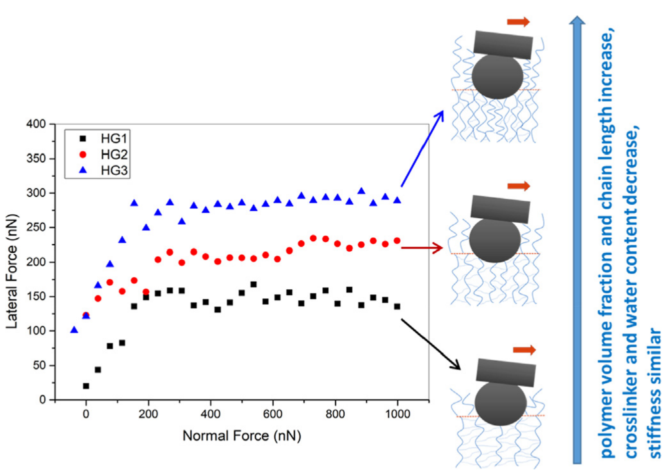
GRAPHICAL ABSTRACT



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ABSTRACT

Hypothesis: The stiffness has been found to regulate hydrogel performances and applications. However, the key interfacial properties of hydrogels, like friction and adhesion are not controlled by the stiffness, but are altered by the structure and composition of hydrogels, like polymer volume fraction and crosslinking degree.

Experiments: Colloidal probe atomic force microscopy has been use to investigate the relationship between tribological properties (friction and adhesion) and composition of hydrogels with similar stiffness, but different polymer volume fractions and crosslinking degrees.

Findings: The interfacial normal and lateral (friction) forces of hydrogels are not directly correlated to the stiffness, but altered by the hydrogel structure and composition. For normal force measurements, the adhesion increases with polymer volume fraction but decreases with crosslinking degree. For lateral force

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measurements, friction increases with polymer volume fraction, but decreases with crosslinking degree. In the low normal force regime, friction is mainly adhesion-controlled and increases significantly with the adhesion and polymer volume fraction. In the high normal force regime, friction is predominantly load-controlled and shows slow increase with normal force.

1. Introduction

Tribological effects, including adhesion and friction, are critical interfacial properties for processes involving contact and relative motion between surfaces [1–4]. The energy efficiency and working life of machinery could be significantly improved by reducing adhesion and friction [4]. Macrotribology has been scientifically investigated ever since 15th century [5–11]. However, only since mid-1990s nanotribology has developed dramatically due to direct measurements of interfacial forces using atomic force microscopy (AFM), particularly with a colloidal probe [8,9,12–15]. The advances in nanotribological studies help to extend the lifetime of nanodevices and also assist in the elucidation of tribology mechanisms due to reduced influences of surface asperities and roughness [16].

Hydrogels are three-dimensional networks composed of crosslinked hydrophilic polymers with typical water contents between 80 and 95 wt% [17,18]. They have properties similar to both solids and liquids. Like solids, hydrogels deform under stress and recover to their original shapes once the stress is released; as for liquids, solutes are able to diffuse in hydrogels if they are smaller than the mesh size of the polymer network [17]. Hydrogels are frequently used as model extracellular matrix (ECM) [19], tissue and organ surfaces [20], and have wide applications in biomedical devices [21], biosensors [22], drug delivery [23,24], tissue engineering [25,26], etc.

Colloid probe AFM lateral force measurements can be used to study the lubricity of the hydrogel/ water interface [27,28]. Adhesion and friction, are of critical importance to hydrogels, as most of hydrogel applications involve contact and relative motion against other surfaces. Investigation of the tribological mechanisms of hydrogels helps to reveal the lubrication mechanism in biological systems and aid the design of novel low-friction and long-lasting artificial tissue and organs. Hydrogel friction has been found to be strongly dependent on the crosslinking degrees of the polymer network [27,29,30], as well as the surface properties of opposing substrates [6,7], and measurement conditions, like normal load [31] and sliding velocity [7,32,33].

For example, Gong et al. found that the interfacial interactions between the hydrogel surface and the opposing surface have significant effects on hydrogel friction. They found that polyelectrolyte hydrogels with the same charges showed very low frictional force when sliding across each other due to electric repulsive forces between the two surfaces; in contrast, two polyelectrolyte hydrogels carrying the opposite charges showed high adhesion and friction during sliding across each other due to electric attractive forces [7]. Gong et al. also found that under repulsive conditions (polymer-surface interaction) friction was mediated by the lubrication of a hydrated layer of polymer chains whereas under adhesive conditions, elastic deformation dominates [34]. Nordgren and Rutland [35] have studied symmetric interactions between end grafted dual responsive polymers on solid surfaces, and showed that as the aqueous solvent conditions are changed from low pH to high pH, the friction increases dramatically due to the transition from extended hydrated chains to collapsed adhesive ones. Espinosa-Marzal et al. investigated the influence of loading conditions and temperature on static friction of hydrogels and found that modulating the hydrogel's mesh size and its surface region could control static friction, contact ageing and thus wear of hydrogels with different Young's modulus [28]. They also investigated the friction hydrogels as a function of sliding velocity; the results revealed the intermittent friction above and below the transition velocity with characteristics depend on the hydrogel network, applied load and sliding velocity [36]. Sawyer et al. measured the lowest friction coefficient at the lowest sliding speeds for their hydrogel systems, which was opposite to the classic Stribeck curve. They associated these distinct regimes with polymer fluctuation at the surface instead of fluid film lubrication [37]. Benetti et al. investigated the effect of crosslinker concentration in the monomer feed on mechanical and tribological behaviour of different polymer hydrogels [27,29]. They found that the Young's moduli and friction coefficients increased with the crosslinking degree due to the reduction of polymer chain conformational freedom.

Stiffness is the resistance of an object to deformation in response to an applied force; it is usually measured by the Young's modulus, which is stress (force per unit area) over strain (proportional deformation) in the linear elasticity regime of a uniaxial deformation. [38] The Young's moduli of hydrogels can be tuned from a few kPa to a few MPa by varying the crosslinking degree of the polymer network [39]. Because of this, hydrogels have been widely used in biomedical applications as ECMs, artificial tissue and organ surfaces, etc.

Recently the stiffness of ECMs made by hydrogels has been found to regulate stem cell differentiation [19,40–42]. However, this work did not consider that the lateral (frictional) interactions a stem cell experiences as it moves across and embeds in, the ECM could be a factor. While the lubricity of hydrogels with different stiffness has been investigated [27,36], how friction changes when the stiffness is similar but the polymer volume fraction and crosslinking degree vary is unknown. In this study, we address this knowledge gap by using colloidal probe AFM to investigate the adhesion and friction of polyacrylamide (PAAM) hydrogels with similar stiffness, towards experiments probing whether friction plays a role in stem cell differentiation.

2. Experimental

2.1. Polyacrylamide (PAAM) hydrogel preparation

PAAM hydrogels were prepared from a hydrogel polymer solution containing acrylamide monomers, crosslinker N,N methylene-bis-acrylamide, 1/100 vol of 10% Ammonium Persulfate (APS), and 1/1000 vol of N,N,N′,N′-Tetramethylethylenediamine (TEMED). Glass coverslips (Fisher), cleaned of organics and oxidized by exposing both sides for 3 min to UV/ozone (BioForce), were immediately functionalized with 20 mM 3-(trimethoxysilyl)propyl methacrylate (Sigma) in ethanol, washed with ethanol, and then dried. 25 μL of the prepared hydrogel solution was sandwiched between a functionalized coverslip and a dichlorodimethylsilane (DCDMS)-treated glass slide to ensure easy detachment of hydrogels subsequent to polymerization. Hydrogels were allowed to polymerize for 15 min then soaked in phosphate-buffered saline (PBS) prior to AFM normal force and friction tests. The obtained three hydrogels had the same stiffness of 30 kPa according to a previous study [19]; the ratios of acrylamide%/bis-acrylamide% were 8/0.55 (HG1), 10/0.3 (HG2), and 20/0.15 (HG3), thus the water contents of the hydrogels are 91.45%, 89.70% and 78.15%, respectively

2.2. AFM colloidal probe preparation

A colloidal probe was prepared by gluing a 5 μm diameter spherical borosilicate particle (Bangs Lab Inc., USA) onto a tipless cantilever (HQ: NSC36/tipless, MikroMasch, Tallinn, Estonia) using a two part epoxy (Selley's Super Strength Araldite). After curing for at least 24 h, cantilevers were imaged using an optical microscope (Zeiss, Axioskop 40) with built-in camera (Zeiss, AxioCam Cc1). Cantilever plane dimensions and the diameter of colloidal probes were measured using AxioVision software (Zeiss, AxioVision 4.7). The normal spring constants were 0.8 ± 0.2 N/m by Sader methods [43].

2.3. AFM measurements

Normal force and friction measurements were performed using a Digital Instruments Dimension 3000 AFM. The silica colloidal probe was cleaned before use by Milli-Q water, dried under nitrogen, and irradiated with ultraviolet light for 20 min to remove organic contaminants. A glass petri dish containing hydrogel samples immersed in PBS at pH 7 was put on the AFM sample disk. Normal force curves were collected by moving the hydrogel surface towards the colloidal probe and detecting the cantilever deflection as a function of separation. The ramp size and rate were 2 μm and 0.2 Hz respectively. Standard methods were used to convert deflection vs separation data to normal force vs apparent separation curves [44]. The Young's modulus was extracted from the retraction force curves using Linearized Model with adhesion force included using Nanoscope Analysis software.

Lateral forces were obtained in contact mode by performing AFM scans with a scan angle of 90° (with respect to the long axis of the cantilever) and with the slow scan axis disabled [44]. The scan size was 5 μm, and scan rate was 6 Hz. The torsional spring constant was calculated using the following equation [45]:

$$k_\phi = k_z \frac{2L^2}{3(1+\nu)} \tag{1}$$

where k_φ is the torsional spring constant, k_z is the normal spring constant, L is the length of the rectangular cantilever, v is the Poisson ratio of the cantilever.

The lateral deflection signal (*i.e.*, cantilever twist) was converted to lateral force using a customized function produced in Matlab which takes into account the torsional spring constant and the geometrical dimensions of the cantilever [46,47]. These experiments were repeated more than three times; the obtained lateral force data showed the same trend with small variations. The repeated normal force curve and lateral force curve data are shown in SI Figs. S1 and S2.

3. Results and discussion

A schematic of the structure of the three hydrogels (HG1, HG2 and HG3, respectively) with different polymer/crosslinker ratios, but similar elastic modulus, is shown in Fig. 1(a). HG1 is made from the hydrogel solution containing the lowest concentration of the acrylamide monomer, but the highest concentration of the crosslinker N,N methylene-bis-acrylamide, thus this hydrogel has the highest crosslinking degree in the bulk, but the shortest and sparsest dangling polymer chains at the interface. In contrast, the original solution for HG3 contains the highest concentration of monomers but the lowest concentration of the crosslinker, therefore HG3 has the lowest crosslinking degree in the bulk but contains the longest and densest surface dangling polymers chains.

3.1. Normal forces

Typical normal force-distance profiles are recorded in the approach (blue) and retraction directions (red) on the three hydrogel surfaces immersed in PBS, as shown in Fig. 2. These curves are fully reproducible during the entire course of the measurements, indicating that the hydrogel surface is not affected by the force-measuring process.

For HG1, as the silica colloidal AFM probe approaches to the hydrogel surface, a tiny repulsive force of ~0.1 nN is observed from a separation of ~1000 nm (Fig. 2(a) and (b)), and increases gently to ~10 nN until the separation is ~200 nm. Then this repulsion increases more sharply and finally reaches a value of ~170 nN at "zero separation", where the normal force increases extremely significantly with separation [48,49]. Since the polyacrylamide polymer chains, the main components of the hydrogel, are not charged, whereas the silica colloidal probe is negatively charged at pH = 7, this two-stage repulsive force detected in the approach direction is likely due to the compression of the dangling polymer chains at large separations and compression of the well crosslinked hydrogel surface at small separations, as shown in the schematic of Fig. 2(b). The lengths of the dangling chains are ~800 nm, which are in the same range with those found by Espinosa-Marzal et al. [28]. For HG2 and HG3, the repulsive forces in approach have the same shape with HG1, first increase gently from a separation of 1000 nm, then increase significantly when the separation is smaller than 200 nm, and finally reach a force of ~200 nN at zero separation. The distance in the first low repulsive force region is generally the same for all three hydrogels, which means the supposed length differences in the dangling polymer chains could not be identified by the silica colloidal AFM probe during normal force measurements. This is because longer dangling polymer chains are partially coiled at the hydrogel interface in PBS at pH 7, thus show no distinguishable length differences with shorter chains or because the initial compression forces for long but less crosslinked polymer chains are too small to be detected.

The retraction data for all three hydrogels shows an adhesion with small steps. A relatively weak adhesion of 7 ± 2 nN is observed on the HG1 surface until the separation is beyond 400 nm; this is likely due to a bridging force originated from the van der Waals and hydrogen bonding interactions between the silica colloid surface and the dangling polymer chains exposed at the hydrogel interface [50–54]. With the increase of polymer volume fraction and decrease of crosslinkers in the hydrogels, the adhesion increases progressively from HG1 (7 ± 2 nN) to HG2 (9 ± 3 nN) and HG3 (16 ± 5 nN), and the small steps are more obvious. These results are similar with Benetti et al.'s study, which showed that less crosslinked polymer hydrogels displayed more significant adhesion owing to increased freedom of polymer chains [27]. The small steps detected in the retraction force curves likely result from the detachment of chains, or ensembles of chains bridging between the dangling polymer chain and the silica probe [27,53,55]; they are the most significant for HG3, consistent with the idea of longer dangling polymer chains leading to longer ranged bridging interactions with the silica colloidal AFM probe surface. The larger magnitude of the force could be related to a stronger interaction, or a greater footprint of the chains on the counter surface, or both.

The Young's moduli extracted from the retraction force curves via Nanoscope Analysis software are 39 ± 5 kPa, 44 ± 5 kPa, and 55 ± 11 kPa for HG1, HG2, and HG3, respectively, which are

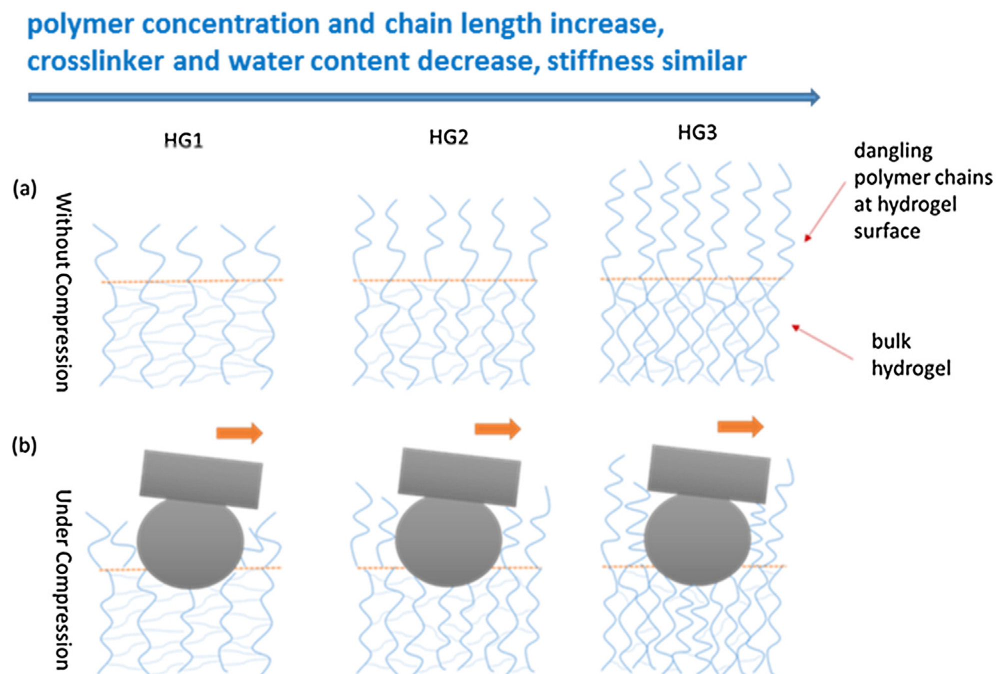

Fig. 1. Schematic of the hydrogel structure: (a) without compression, and (b) under compression for acrylamide 8%/bis-acrylamide 0.55% (HG1), acrylamide 10%/bis-acrylamide 0.30% (HG2), and acrylamide 20%/bis-acrylamide 0.18% (HG3).

comparable, but slightly higher than the values (30 kPa) published in previous study [19]. This variation is likely because previous study used a gentle indentation force [19], thus just indented the soft dangling polymer chains; whereas a higher force was applied in our study and the AFM probe indented into the well crosslinked bulk hydrogels, thus showed higher Young's moduli values. Similar effects have been found by Espinosa-Marzal et al. [28].

3.2. Lateral forces

Fig. 3 shows the lateral force versus the normal load for a silica colloidal AFM probe sliding on three hydrogel surfaces immersed in PBS at the same sliding speed of 60 μm/s when the normal load increases from zero nN to 1000 nN. The lateral force versus the normal load when the normal load value decreases is shown in Fig. S2. No significant differences were found between the loading and unloading curves, indicating no obvious restructuring or work hardening at the hydrogel interface during the friction measurements. Two regions are present in the lateral force data for all three hydrogels investigated: an initial sharp increase region at the low normal forces (F_N < 200 nN) and a gentle increase/flat region at the high normal forces (F_N > 200 nN), similar to those found on graphite surfaces [56].

For HG1, when the normal force is below 200 nN, the lateral forces increases significantly from 15 nN to 150 nN; however, as the normal force continues to increase up to 1000 nN, the lateral force remains almost constant. For HG2 and HG3, the lateral forces also increase significantly at low normal forces (<200 nN), from ~50 nN and ~100 nN to 200 nN and 300 nN, respectively, and then increase more gently.

The friction results are generally in line with the normal force-distance profiles in the approach direction (the blue curves shown in Fig. 2). For all three hydrogel surfaces, the normal forces (170–200 nN) at the zero separation (Fig. 2) are consistent with the critical normal forces observed in friction measurements (Fig. 3), where the lateral force starts to increase gently.

According to the modified Amontons' law [1], the total lateral force (F_{LT}) required for one surface to slide across another could be described by:

$$F_{LT} = S_c \pi a^2 + \mu F_N \quad (2)$$

where S_c is the shear strength, a is the surface contact radius, μ is the friction coefficient, F_N is the normal force. The first term is adhesion-controlled; its magnitude is governed by both surface roughness and the physiochemical interactions between surfaces. The second term is load-controlled; its magnitude is dictated predominately by surface topography and roughness.

Lateral force is mainly load-controlled for a rigid surface; the surface contact radius (a) is small and almost constant with the increase of normal force [1]. However, on a soft surface including hydrogel, lateral force, especially in the initial low normal force region, is significantly influenced by adhesion [17,57,58]. An obvious evidence is that none of the lateral forces are zero at zero normal forces for all three hydrogel in this study, as shown in Fig. 3. The adhesion can be seen as an additional load to the externally applied load and the intercept of the data on the normal force axis (at a negative value) can be considered a measure of the adhesion, or "contact adhesion". As the same with the normal force experiments the largest value of this "contact adhesion"[59] is observed for HG3 and the lowest for HG1 though the absolute values are not the same. The nature of the adhesion measurement is rather different: the "contact adhesion" measurement is obtained from a sliding measurement while in contact, whereas a normal force measurement is dynamic, and for polymeric systems the magnitude of the force depends on the time in contact and rate of separation [53,55,59,60].

The surface contact radius (a) in the adhesion-controlled term could be coarsely estimated using different contact mechanics models. The Hertz model only takes into consideration of hard wall repulsion at contact without considering attractive forces at all [61]. The Derjaguin–Müller–Toporov (DMT) model is used for rigid materials that have lower surface energy and are resistant to deformation [62]. The Johnson–Kendall–Roberts (JKR) model is applied

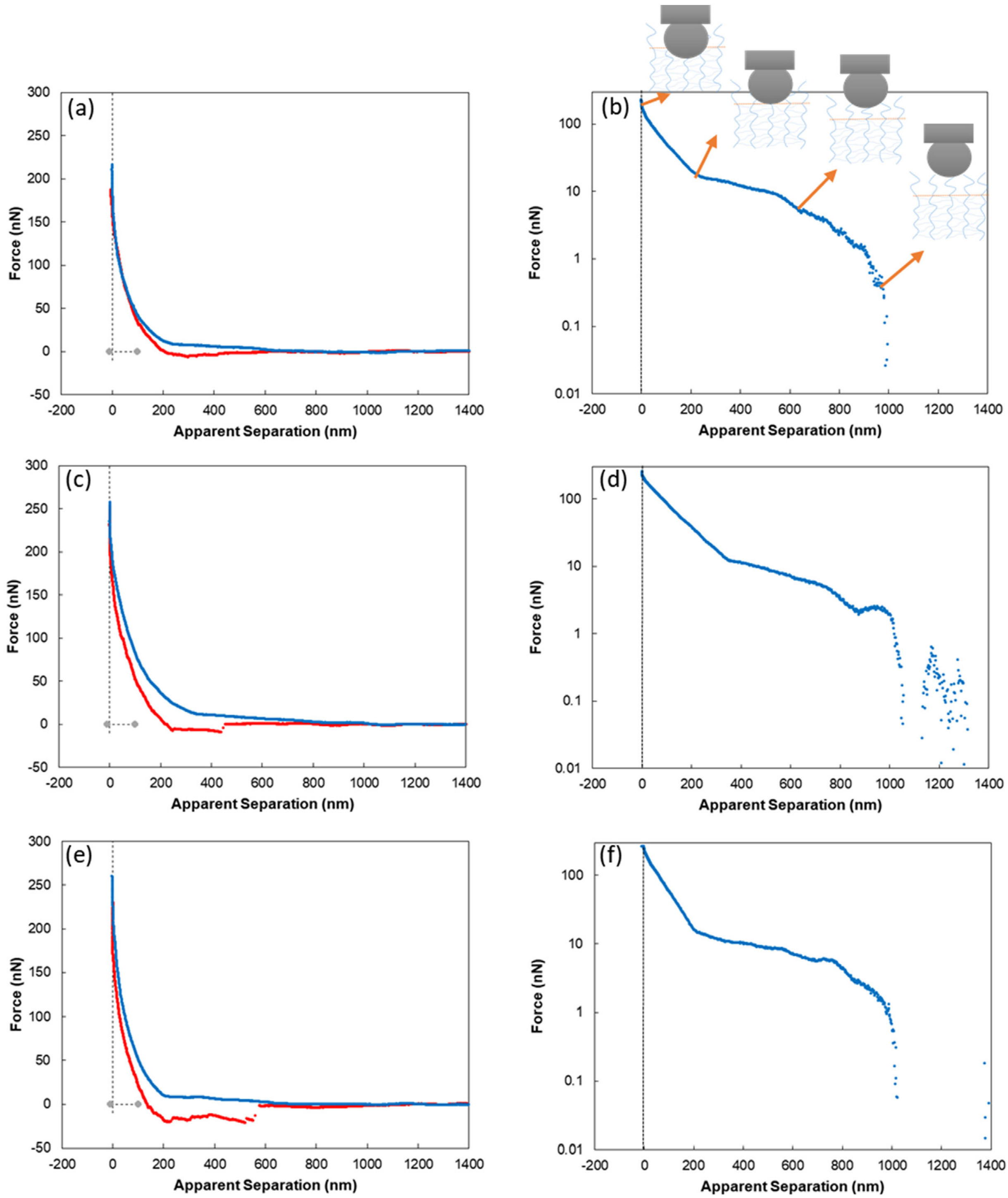

Fig. 2. Typical normal force versus apparent separation profile for a silica colloidal AFM probe approaching (blue) and retracting (red) from different hydrogel surfaces immersed in PBS for normal (a, c, e) and log (b, d, f) Y axis scales: acrylamide 8%/bis-acrylamide 0.55% (HG1) (a, b), acrylamide 10%/bis-acrylamide 0.30% (HG2) (c, d), and acrylamide 20%/bis-acrylamide 0.18% (HG3) (e, f). The gray dots in the figures indicate zero at x and y axes.

to describe soft materials with high surface energy and large deformation [63]. Therefore, we used JKR model in this study to estimate the surface contact radii, as shown in Table 1. The details on how to achieve the surface contact radii are described in SI. At zero normal force, the surface contact radius increases slightly from HG1 to HG3, which has the same trend with the increase of the adhesion, meaning that higher adhesion leading to larger surface contact radius.

When the normal force is very small, the contribution of adhesion is dominant, thus the load-controlled term in Eq. (2) is negligible, and the shear strength, S_c, could be extracted from $F_T = S_c \pi a^2$ [64], as listed in Table 2. S_c is mainly determined by the interactions between the two contact surfaces [1]. The obtained S_c is in the same order of the Young's modulus, and increases with the polymer volume fraction while decreases with crosslinking degree from HG1 to HG3, similar as seen by Pitenis and Sawyer [65]. Silica

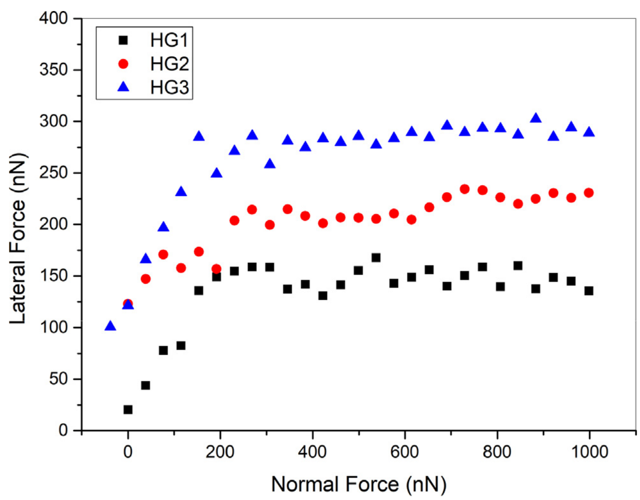

Fig. 3. Lateral force versus normal load in PBS for a silica colloidal probe sliding on three hydrogel surfaces at a sliding speed of 60 μm/s.

Table 1
The surface contact radii of hydrogels at normal forces of 0 nN and 200 nN, respectively. The errors are within 10%.

Normal Force	HG1	HG2	HG3
0 nN (μm)	1.0	1.1	1.3
200 nN (μm)	2.2	2.3	2.3

Table 2
The shear strength in the low normal force regime obtained from JKR fitting and the friction coefficient in the high normal force regime.

	HG1	HG2	HG3
Shear Strength, τ (KPa)	5.5	13.9	14.4
Friction coefficient, μ	0.008	0.04	0.04

has been found form H-bonds with PAAM hydrogels via Si—O of silica with the NH_2 group of the polymer chains. Longer and less cross-linked polymer chains contain more sites to form H-bonds with the silica colloidal AFM probe. Our results here suggest that higher polymer volume fraction and lower crosslinking degree lead to stronger H-bonding effects via more binding sites of the hydrogel polymer chains with the silica colloidal AFM probe during sliding, thus result in higher shear strength, similar as seen before at graphene and graphene oxide interfaces [66,67].

At the critical normal load of 200 nN, the increase in surface contact radius is not significant and the values are all close to 2.5 μm, *i.e.*, the radius of the silica colloidal AFM probe and the maximum contact radius. This means the effect of adhesion is less significant with the increase of normal force. When the normal force is higher than 200 nN, the surface contact radius reaches the saturation value and could not increase significantly, thus the friction in the high normal force regime (F_N > 200 nN) is mainly load-controlled and determined by the surface roughness. The friction coefficients, the gradients of lateral force over normal force, for all three hydrogels are very small in this high normal force regime, meaning that the hydrogel surfaces are all very smooth and show low friction coefficients under such conditions. However, the friction coefficients for HG2 and HG3 are slightly higher than that of HG1, probably because HG 2 and HG 3 are slightly rougher due to larger pore sizes as found by Wen et al. [19] Also the dehydrations of polymer chains for HG2 and HG3 are more significant during compressing, leading to more reduced fluid lubricants within the hydrogel and thus relatively higher friction, similar as seen in other studies [27,68].

In previous work Benetti et al. found the hydrogel friction increases with the crosslinking degree for the same polymer volume fraction [27]. They attributed this to energy dissipation being mainly due to conformational changes of the polymer chains, which are determined by the crosslinking degree. Higher crosslinking degree leads to less conformational freedom and reduced fluid lubrication within the film, therefore results in higher friction [27,51]. In our case, while the stiffness of the hydrogel is kept constant, both the polymer volume fractions and crosslinking degrees are different. In this work, although the decrease of crosslinking degree leads to reduced dissipation via chain conformation change; the friction still increases with polymer volume fraction due to increased H-bonding sites between the hydrogel and the silica probe. This means for these equally stiff hydrogels with different polymer volume fraction and crosslinking degree, energy dissipation is predominantly via shear between polymer chains and silica probe, and chain conformation dissipation is less important. This does not mean the prior work is wrong, but rather that both the hydrogel polymer concentration and crosslinking degree are important for friction in hydrogel systems.

4. Conclusion

Previous work has suggested that for the hydrogels with similar polymer volume fraction but different crosslinking degree, the lateral force increases with the crosslinking degree due to reduced conformational freedom and fluid lubrication at the hydrogel interface [27,29,51]. However, our colloidal probe AFM measurements reveal that the situation is more complicated and lateral force increases with polymer volume fraction, but decreases with crosslinking degree, over the entire normal force regime investigated. Adhesion increases with polymer volume fraction due to stronger bridging effects caused by van der Waals and hydrogen bonding interactions between chains and the silica sphere [50–54], and decreases with crosslinking degree owing to reduced polymer chain freedom [27].

The results from this study indicate that the key interfacial properties of hydrogels, namely adhesion and friction are not directly related to the stiffness; they are more influenced by hydrogel structure and composition. Recent studies have shown that stem cell differentiation is regulated by the stiffness of the hydrogel substrates [19,40–42]; however, the interactions, *i.e.* adhesion and friction between stem cells and hydrogel substrates could also have effects, but have been rarely studied. Future experiments will examine the implications of the findings in current work for stem cell differentiation.

CRediT authorship contribution statement

Hua Li: conceptualization, data curation, formal analysis, investigation, methodology, Writing - original draft, Writing - review & editing. **Yu Suk Choi**: conceptualization and methodology. **Mark W. Rutland**: formal analysis, investigation, methodology, Writing - review & editing. **Rob Atkin**: conceptualization, methodology, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcis.2019.12.045.

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